Tranched CDX Trading

Seminar Module

conducted by

ACF Consultants Ltd





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Topics Covered

- Introduction to tranched CDXs
- Tranched CDX example
- Correlation and tranched CDXs
- Investing in tranches
- Trading tranches
- The concept of delta
- Delta-neutral correlation trades
- Base correlation
- Risk management of CDX tranches
- What went wrong in May 2005

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What is a Tranched CDX?

A tranched CDX is where the assets created represent specific risk tranches of the underlying portfolio, e.g.

- **0%-3%** where investors bear the first 3% of losses on the underlying pool
- **3%-7%** where investors bear the next 4% of losses
- **7%-10%** where investors bear the next 3% of losses
- etc.

These percentages are called the

Attachment and detachment points

The risk-return characteristics of each tranche are affected by:

- **Tranche size** – how wide each tranche is
- **Subordination** – how many tranches are below the current tranche
- **Default probability** – the likelihood of default of each asset in the underlying pool
- **Recovery rates** – the financial impact of a credit event
- **Correlation** – the correlation between defaults of each asset in the underlying pool

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Tranched CDXs and Single Tranches

- Tranched CDXs can be created from an actual asset portfolio
  - Investment bank actually purchases 100 bonds of $5m face value each to create a $500m pool
  - Investment bank must then sell all the tranches to investors
- Single Tranches can also be created synthetically from a reference portfolio
  - Investment bank decides to create a synthetic portfolio of $500m notional size
  - Composition of portfolio references actual bonds, e.g. 100 named bonds of $5m face value each…
  - … but investment bank does not actually purchase these bonds
  - As underlying portfolio is not purchased, investment bank does not need to sell all the tranches
- Example of single-tranche CDX
  - Bank sells $80m investment in the 3%-7% single tranche
  - This references a $2bn synthetic underlying portfolio
  - … because $80m / 4% = $2bn

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CDX Example

To see how a tranched CDX works, let's consider this example:

Asset pool	
Number of names:	100
Notional per name:	$1m
Credit rating of names:	BBB
Default probability:	1% p.a.
Recovery rate:	40%

CDX capital structure	
Notional:	$100m
Equity tranche:	0-3%
Mezzanine tranches:	3-7%
	7-10%
	10-15%
Senior tranche:	15-30%
Super-senior tranche:	30-100%

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CDX Example – Creating the Tranches

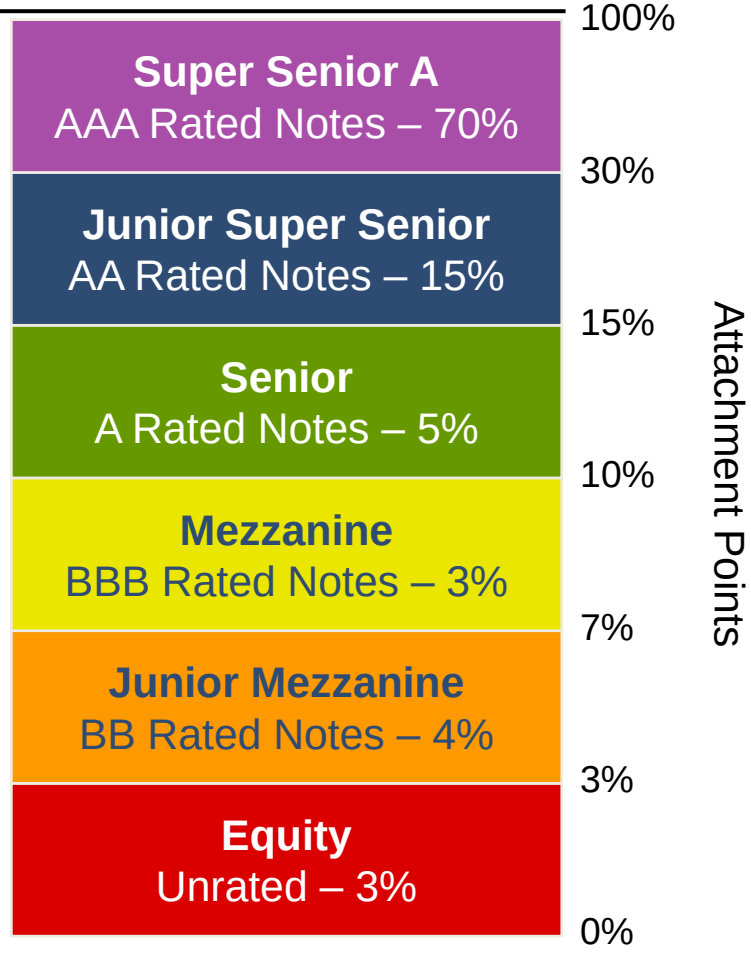
Asset portfolio

$100m Asset Portfolio

100 names
$1m each name

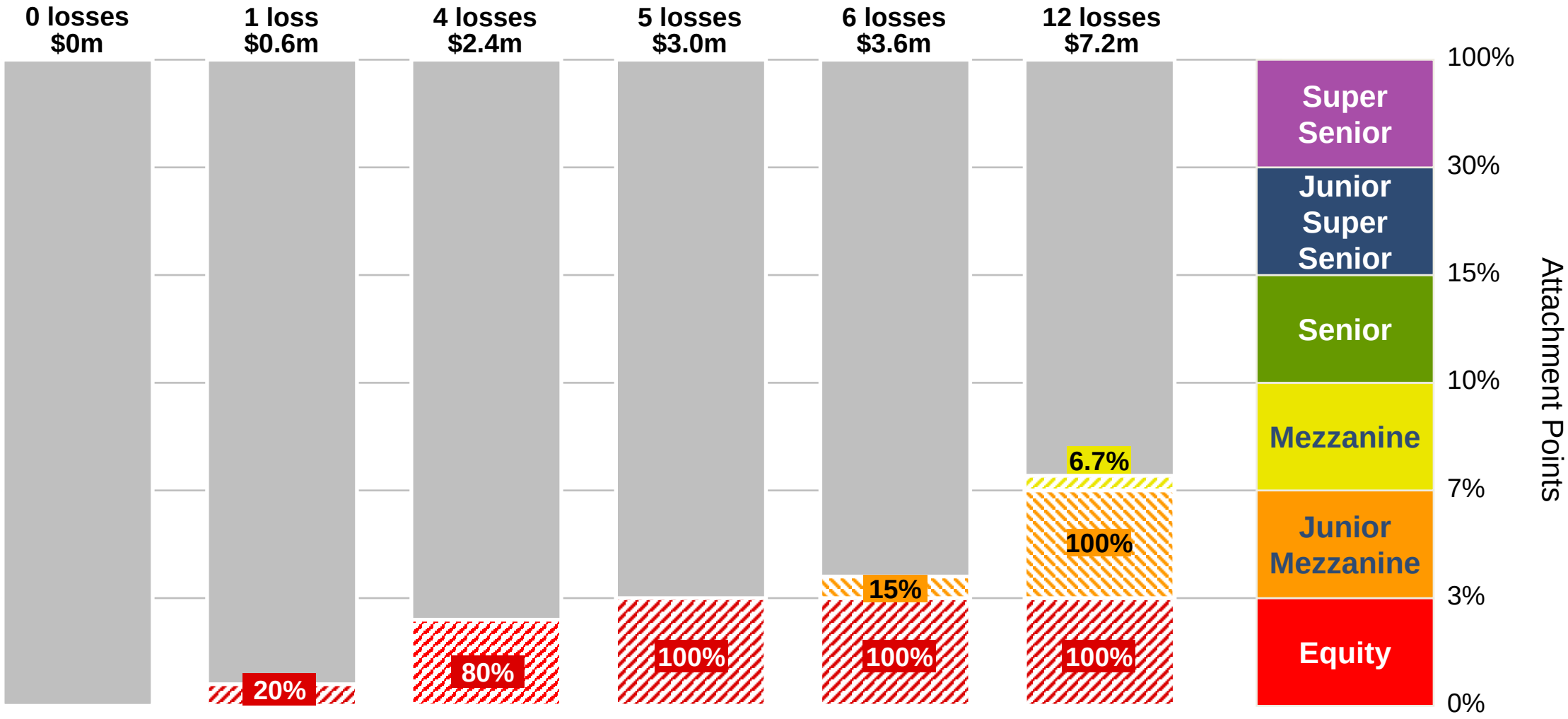


CDX tranche structure



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CDX Example – Impact of Losses



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Tranche Loss Analysis – %age of Portfolio

Number of Losses	Gross Loss	Net Loss	Tranches				
			0%-3%	3%-7%	7%-10%	10%-15%	15%-30%
0	$0	$0	0.0%	0.0%	0.0%	0.0%	0.0%
1	$1,000,000	$600,000	0.6%	0.0%	0.0%	0.0%	0.0%
2	$2,000,000	$1,200,000	1.2%	0.0%	0.0%	0.0%	0.0%
3	$3,000,000	$1,800,000	1.8%	0.0%	0.0%	0.0%	0.0%
4	$4,000,000	$2,400,000	2.4%	0.0%	0.0%	0.0%	0.0%
5	$5,000,000	$3,000,000	3.0%	0.0%	0.0%	0.0%	0.0%
6	$6,000,000	$3,600,000	3.0%	0.6%	0.0%	0.0%	0.0%
7	$7,000,000	$4,200,000	3.0%	1.2%	0.0%	0.0%	0.0%
8	$8,000,000	$4,800,000	3.0%	1.8%	0.0%	0.0%	0.0%
9	$9,000,000	$5,400,000	3.0%	2.4%	0.0%	0.0%	0.0%
10	$10,000,000	$6,000,000	3.0%	3.0%	0.0%	0.0%	0.0%
11	$11,000,000	$6,600,000	3.0%	3.6%	0.0%	0.0%	0.0%
12	$12,000,000	$7,200,000	3.0%	4.0%	0.2%	0.0%	0.0%
13	$13,000,000	$7,800,000	3.0%	4.0%	0.8%	0.0%	0.0%
14	$14,000,000	$8,400,000	3.0%	4.0%	1.4%	0.0%	0.0%
15	$15,000,000	$9,000,000	3.0%	4.0%	2.0%	0.0%	0.0%
16	$16,000,000	$9,600,000	3.0%	4.0%	2.6%	0.0%	0.0%
17	$17,000,000	$10,200,000	3.0%	4.0%	3.0%	0.2%	0.0%

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Tranche Loss Analysis – %age of Tranche

Number of Losses	Gross Loss	Net Loss	Tranches				
			0%-3%	3%-7%	7%-10%	10%-15%	15%-30%
0	$0	$0	0.0%	0.0%	0.0%	0.0%	0.0%
1	$1,000,000	$600,000	20.0%	0.0%	0.0%	0.0%	0.0%
2	$2,000,000	$1,200,000	40.0%	0.0%	0.0%	0.0%	0.0%
3	$3,000,000	$1,800,000	60.0%	0.0%	0.0%	0.0%	0.0%
4	$4,000,000	$2,400,000	80.0%	0.0%	0.0%	0.0%	0.0%
5	$5,000,000	$3,000,000	100.0%	0.0%	0.0%	0.0%	0.0%
6	$6,000,000	$3,600,000	100.0%	15.0%	0.0%	0.0%	0.0%
7	$7,000,000	$4,200,000	100.0%	30.0%	0.0%	0.0%	0.0%
8	$8,000,000	$4,800,000	100.0%	45.0%	0.0%	0.0%	0.0%
9	$9,000,000	$5,400,000	100.0%	60.0%	0.0%	0.0%	0.0%
10	$10,000,000	$6,000,000	100.0%	75.0%	0.0%	0.0%	0.0%
11	$11,000,000	$6,600,000	100.0%	90.0%	0.0%	0.0%	0.0%
12	$12,000,000	$7,200,000	100.0%	100.0%	6.7%	0.0%	0.0%
13	$13,000,000	$7,800,000	100.0%	100.0%	26.7%	0.0%	0.0%
14	$14,000,000	$8,400,000	100.0%	100.0%	46.7%	0.0%	0.0%
15	$15,000,000	$9,000,000	100.0%	100.0%	66.7%	0.0%	0.0%
16	$16,000,000	$9,600,000	100.0%	100.0%	86.7%	0.0%	0.0%
17	$17,000,000	$10,200,000	100.0%	100.0%	100.0%	4.0%	0.0%

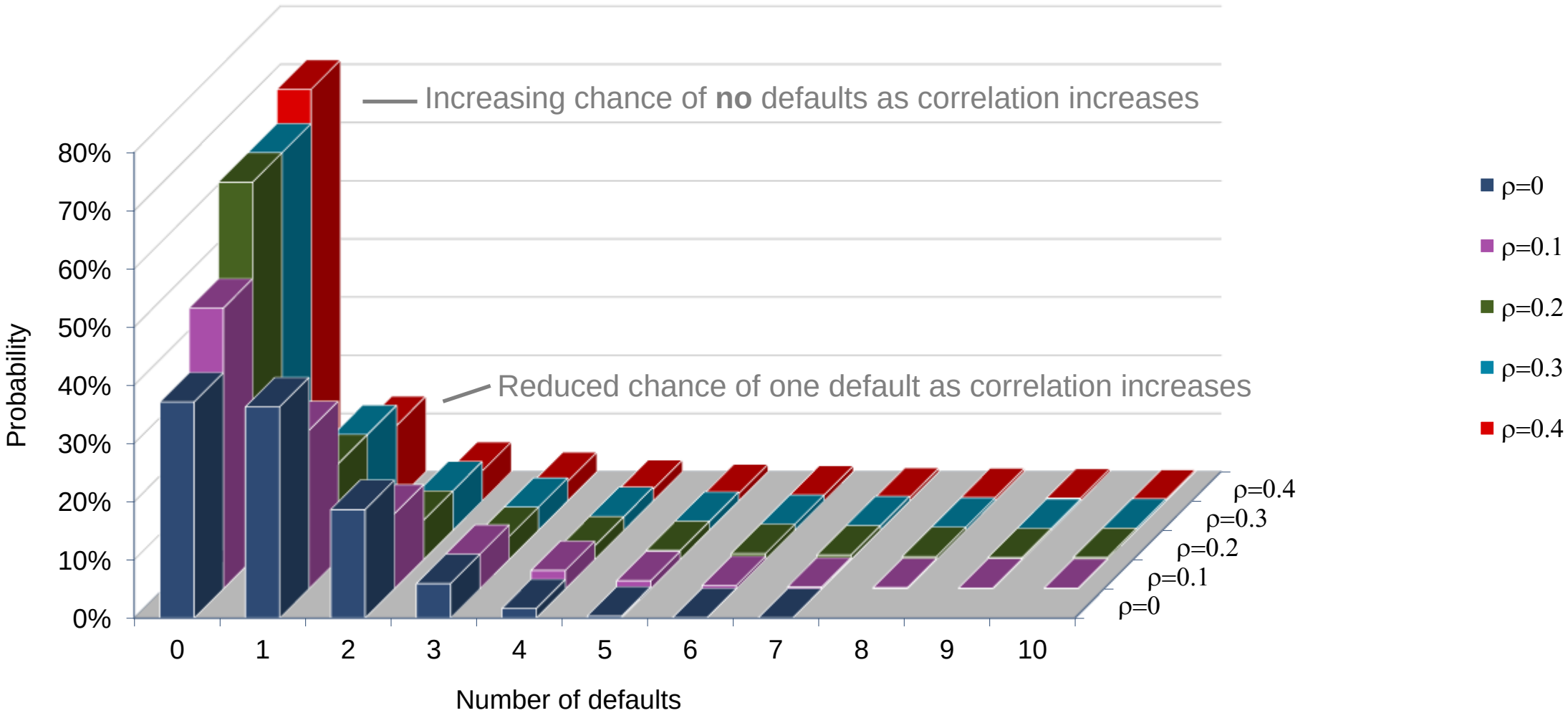
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Correlation and CDXs

- Recall that the risk / return characteristics of CDX tranches depend on:
  - Tranche size
  - Subordination
  - The probability of default of the individual names in the basket
  - The default correlation between individual names
- Higher default probability will increase losses and lower returns for **all** tranches
- Higher default correlation will have different effects for different tranches
  - Mezzanine (and senior) tranches will incur greater losses
  - Equity tranches will incur smaller losses!
- Let's look at the results of a Monte Carlo simulation to study this further...

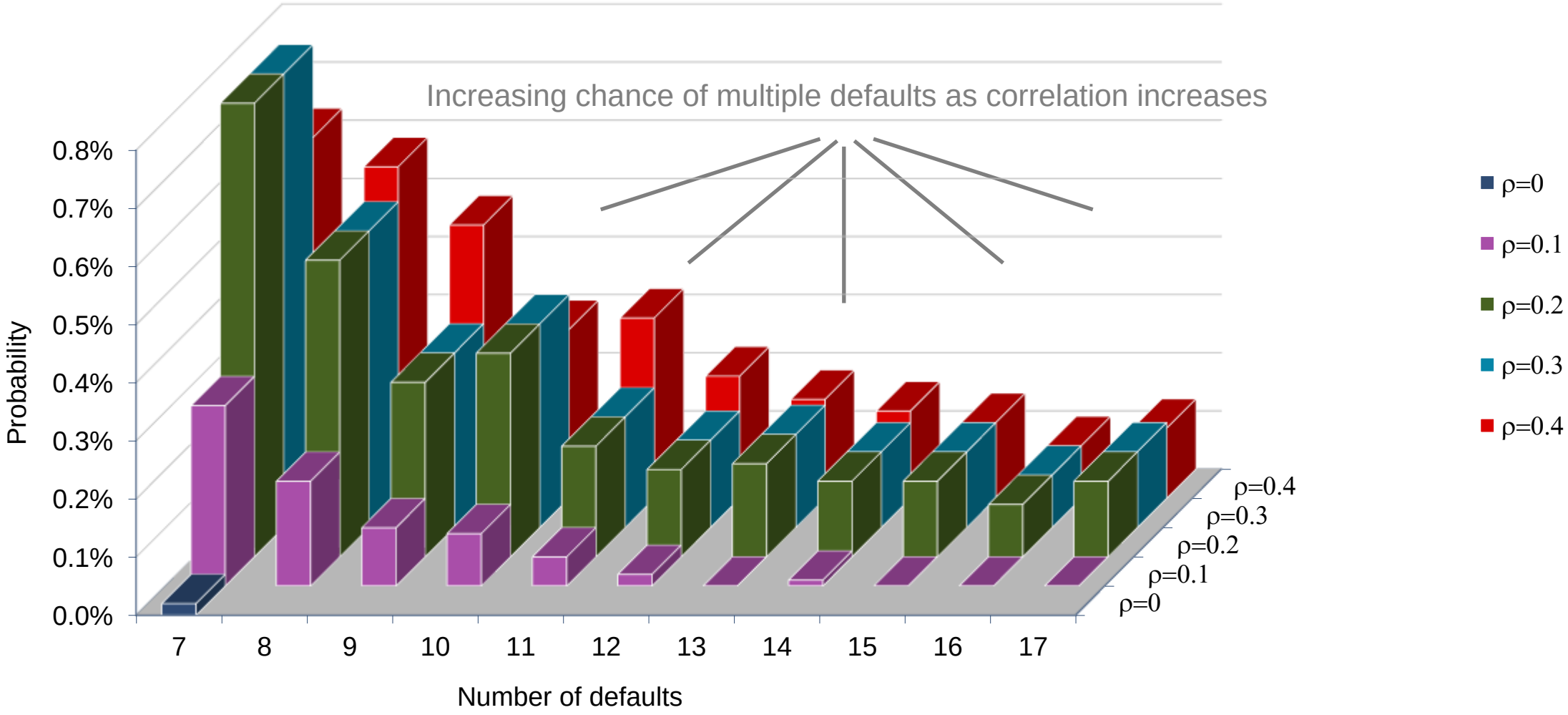
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Effect of Correlation on Portfolio Losses



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Effect of Correlation on Portfolio Losses (Distribution Tail)



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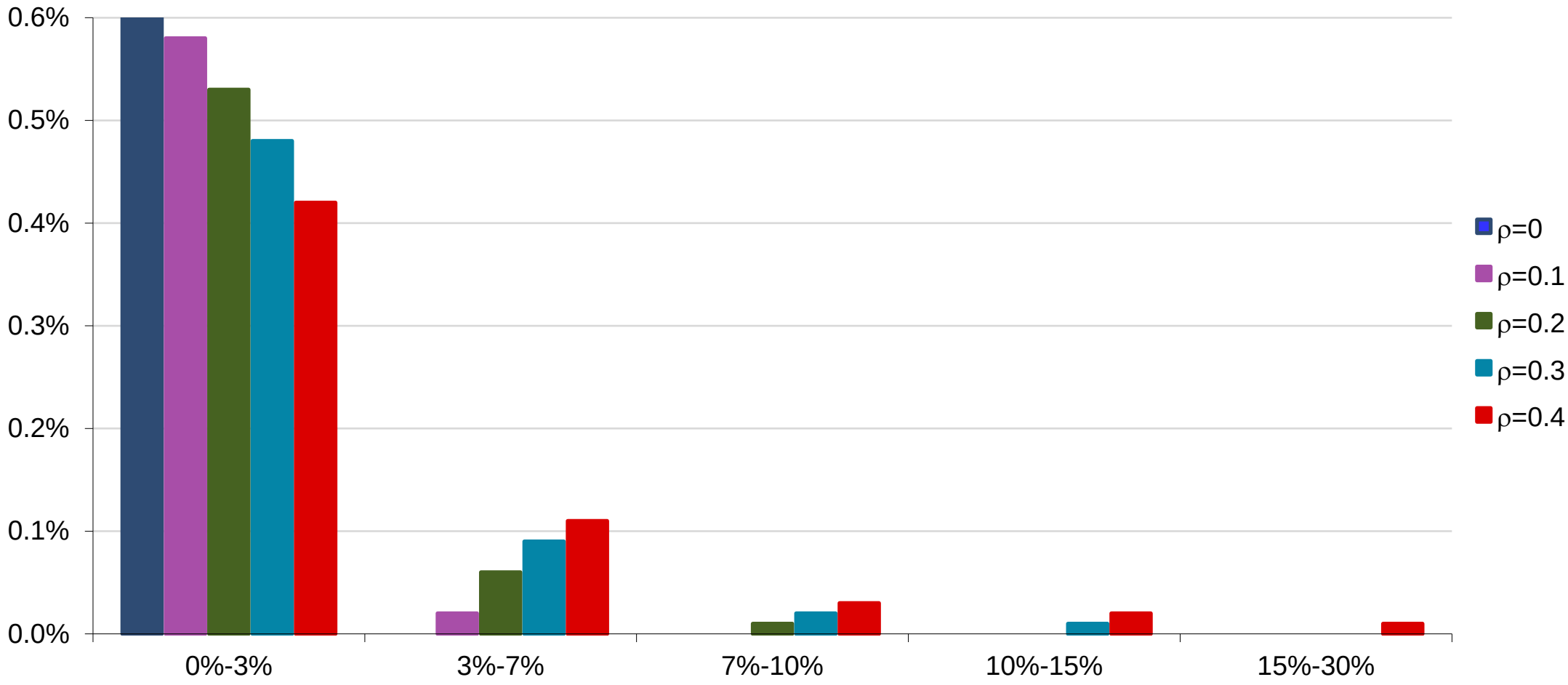
Effect of Correlation – Analysis

As the charts on the previous pages show…

- With zero correlation, there is virtually no chance of multiple defaults in pool:
  - 6 defaults* 1 in 2,000 (* first point at which mezzanine tranche incurs loss)
  - 10 defaults 1 in 14 million
  - 15 defaults 1 in 9 trillion
- In this case, the mezzanine and senior tranches are virtually risk-free
- Expected loss on Mezzanine Tranche E (3%-7%) is only 0.02%
- But with correlation ρ =0.3, the chance of multiple defaults becomes significant:
  - 6 defaults* 1 in 100
  - 10 defaults 1 in 300
  - 15 defaults 1 in 800
- Mezzanine tranches are now considerably more risky
- Expected loss on Mezzanine Tranche E (3%-7%) is now 2.27%!

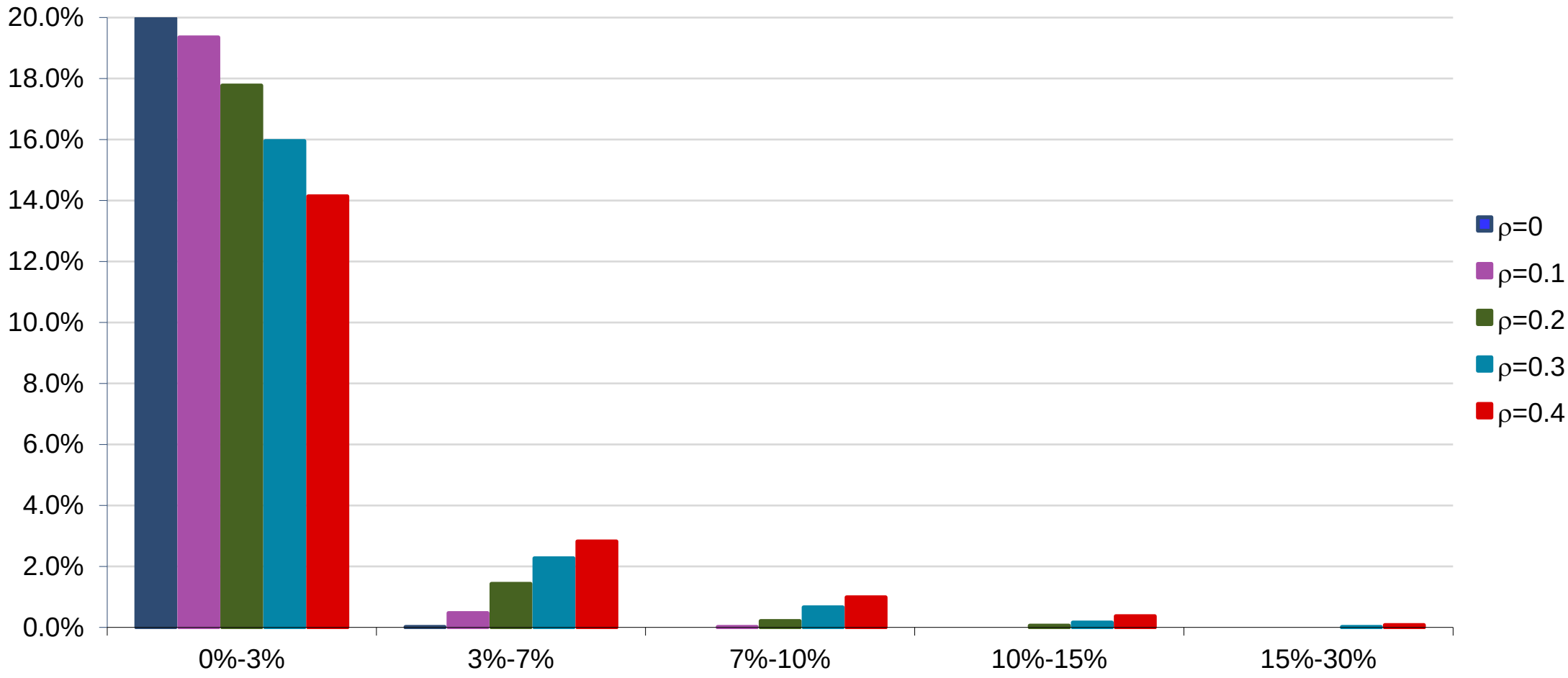
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Portfolio Loss Distribution – All Tranches



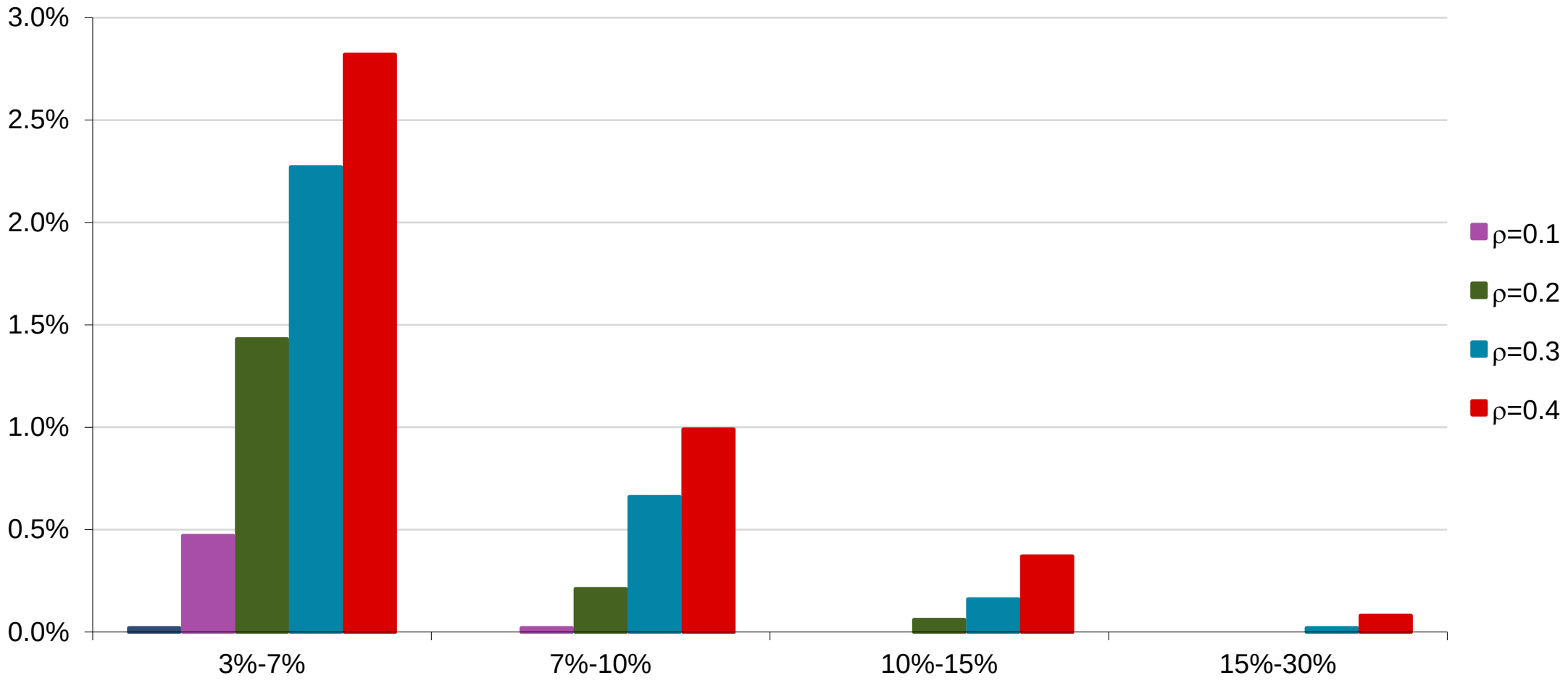
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Tranche Loss Distribution – All Tranches



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Tranche Loss Distribution – Mezzanine Tranches



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Tranche Losses – The Numbers

- The table below shows the **expected losses** for each tranche under different correlation assumptions
- Example: with $\rho = 0.2$
  - Equity tranche expects to lose 17.77%
  - Mezzanine tranche expects to lose 1.43%

Tranche	$\rho = 0$	$\rho = 0.1$	$\rho = 0.2$	$\rho = 0.3$	$\rho = 0.4$
0%-3%	19.98%	19.35%	17.77%	15.95%	14.14%
3%-7%	0.02%	0.47%	1.43%	2.27%	2.82%
7%-10%	0.00%	0.02%	0.21%	0.66%	0.99%
10%-15%	0.00%	0.00%	0.06%	0.16%	0.37%
15%-30%	0.00%	0.00%	0.00%	0.02%	0.08%



Why would anyone want to invest in the equity tranche if they expected to lose 17.77%?

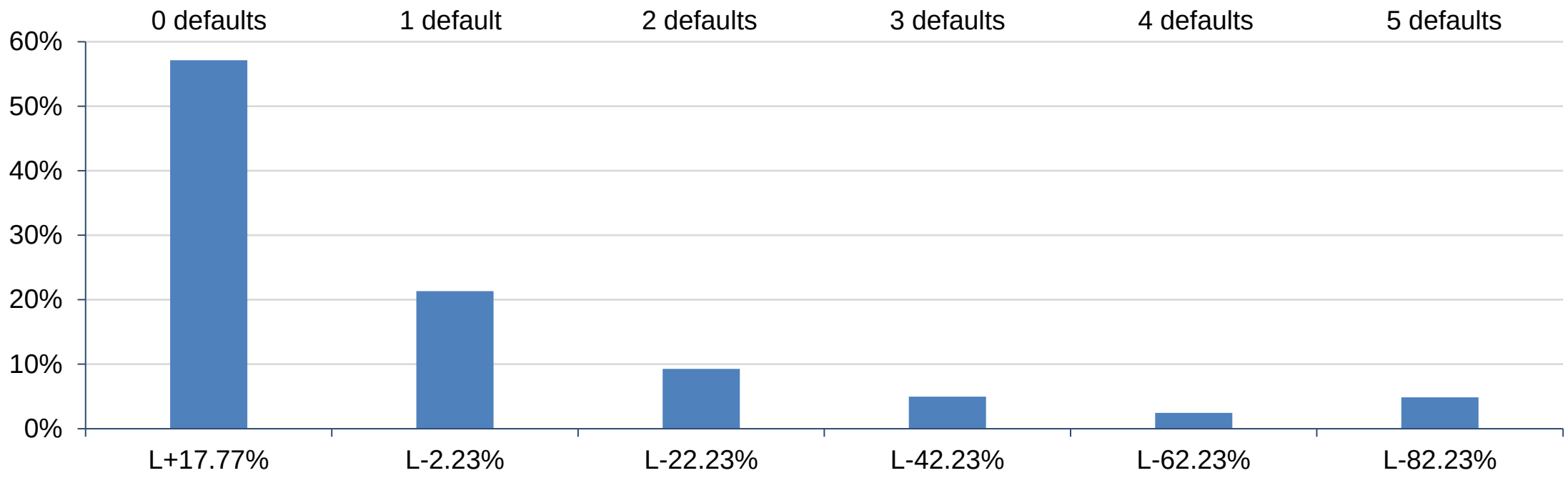
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Investing in Tranches

Q: Why would anyone want to invest in the equity tranche if they expected to lose 17.77%?

A: If they were promised returns of at least L+17.77%, then this would cover expected losses.

Profit / loss distribution for the equity tranche*



** Assumes return of L+17.77%*

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Tranche Performance Statistics – Fair Returns

If an investor received for each tranche a return above LIBOR **equal to** the expected losses, the performance of the investments would be as follows:

Tranche	Gross Return	Net Return	Std Dev	Skewness	Kurtosis	Worst Result	Best Result
0%-3%	L+17.77%	L+ 0bp	27.19%	-1.7	5.2	L-82.23%	L+17.77%
3%-7%	L+143bp	L+ 0bp	9.51%	-8.0	71.2	L-98.57%	L+1.43%
7%-10%	L+21bp	L+ 0bp	4.02%	-21.8	504.2	L-99.79%	L+0.21%
10%-15%	L+ 6bp	L+ 0bp	2.06%	-41.6	1,842.9	L-99.94%	L+0.06%
15%-30%	L+ 0bp	L+ 0bp	0.20%	-95.0	9,263.1	L-20.00%	L+0.00%

Note that the mezzanine tranches have reduced risk as measured by standard deviation, but very much increased risk as measured by negative skewness and kurtosis.

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Tranche Performance Statistics – Market Returns

If an investor received for each tranche a return above LIBOR **greater than** the expected losses – to compensate for risk – the performance of the investments might then be as follows:

Tranche	Gross Return	Net Return	Std Dev	Return / Risk
0%-3%	L+30.00%	L+12.23%	27.19%	0.45
3%-7%	L+180bp	L+37bp	9.51%	0.04
7%-10%	L+50bp	L+ 29bp	4.02%	0.07
10%-15%	L+25bp	L+19bp	2.06%	0.09
15%-30%	L+10bp	L+10bp	0.20%	0.48

These higher returns allow for a reasonable profit margin to be earned. Note that the first two mezzanine tranches still exhibit considerable risk in relation to the returns that can be earned.

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Example of Actual Market Quote

The following table illustrates some actual market quotations:

Tranche	Tranche Size	Bid	Offer	Delta
0%-3%	3%	30 5/8 + 500	31 1/8 + 500	1,700%
3%-7%	4%	180	183	700%
7%-10%	3%	55.5	58.5	250%
10%-15%	5%	23	24.5	125%
15%-30%	15%	7.5	9.5	20%

Note that the equity tranche is quoted ‘points up-front’, while the other tranches are quoted as a running yield.

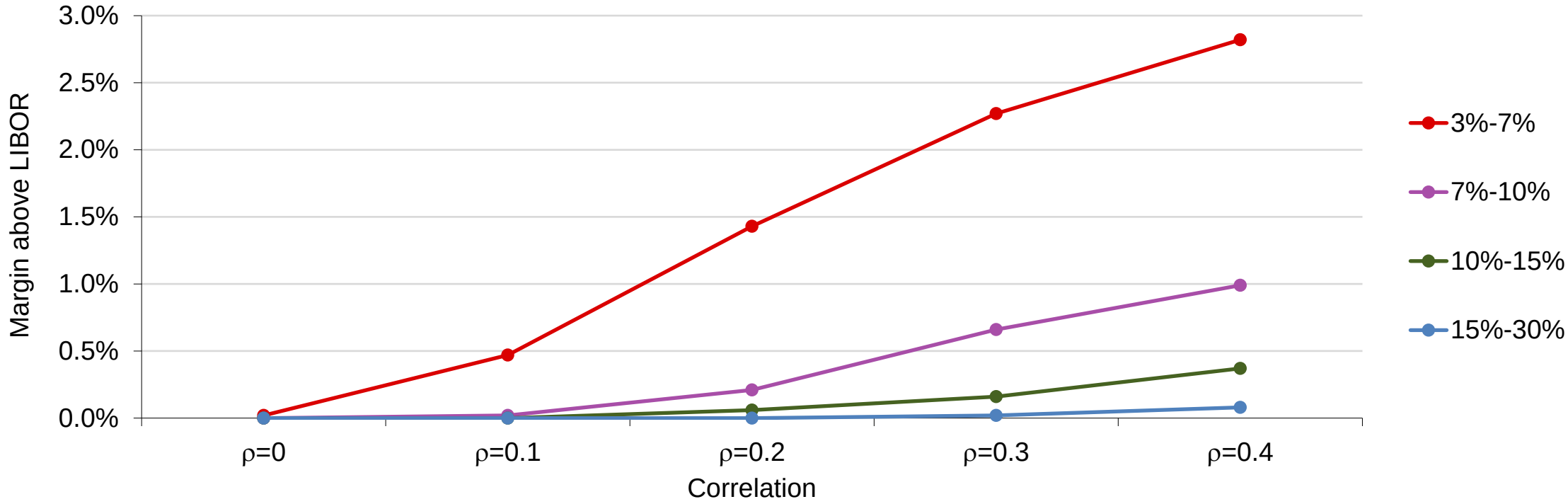
Source: Goldman Sachs
Note: CDX.NA.IG 46 bp per year

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Fair Returns and Correlation

The chart below plots theoretical fair margins against correlation for the mezzanine tranches:

LIBOR margins for mezzanine tranches

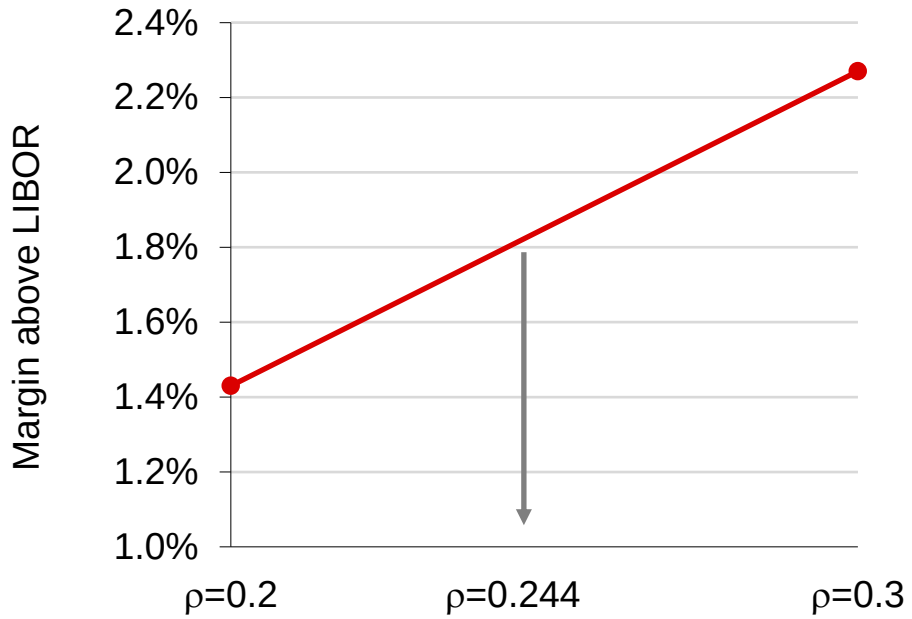


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Implied Correlation

- From the chart on the previous page
  - For ρ = 0.2, fair margin above LIBOR for 3% to 7% tranche is 1.43%
  - For ρ = 0.3, fair margin above LIBOR for 3% to 7% tranche is 2.27%
- If the market is quoting a return of L+1.80%, what is the implied correlation?
- By interpolation, the implied correlation is:

$$\rho = 0.2 + \frac{1.80 - 1.43}{2.27 - 1.43} \times (0.3 - 0.2) = 0.244$$



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Taking a View on Correlation

- Investors wishing to take a view on default correlation can trade CDX tranches
- Suppose the market is pricing a correlation of 0.3
  - Junior Mezzanine (3%-7%) would be priced at fair margin of: L+2.27%
  - Mezzanine (7%-10%) would be priced at fair margin of: L+0.66%
- If an investor believes that correlation will actually be closer to 0.2, we expect CDX Tranche prices to be:
  - Junior Mezzanine (3%-7%) would be priced at fair margin of: L+1.43%
  - Mezzanine (7%-10%) would be priced at fair margin of: L+0.21%
- Buying either of these mezzanine tranches at their original quoted returns would be profitable, because
  - Market returns would fall and the price of the CDX tranche will rise, or
  - Default experience would be lower than expected

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Trading Correlation

- Suppose the investor decided to buy either the 3%-7% or the 7%-10% tranche
- At a market correlation of 0.3, these would be priced at:
  - Junior Mezzanine (3%-7%) would be priced at fair margin of: L+2.27%
  - Mezzanine (7%-10%) would be priced at fair margin of: L+0.66%
- If correlation actually turned out to be 0.2 the investor would make money:

Tranche	Original Return	Final Return	Mean Profit	Std Dev	Return / Risk
3%-7%	L+2.27%	L+1.43%	+0.84%	12.82%	0.07
7%-10%	L+0.66%	L+0.21%	+0.45%	7.36%	0.06

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What if Default Probability Changes?

- Suppose the investor was right about correlation actually being 0.2
- The investor buys either the 3%-7% or the 7%-10% tranche
- But let's suppose the default probability increases from 1.0% to 1.5%
- Even though the investor is right about correlation, he now makes a loss on the 3% -7% tranche, and only a tiny profit on the 7%-10% tranche:

Tranche	Original Return	Final Return	Mean Profit	Std Dev	Return / Risk
3%-7%	L+2.27%	L+2.93%	−0.66%	13.81%	−0.05
7%-10%	L+0.66%	L+0.56%	+0.10%	6.81%	0.01

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The Concept of Delta

- Using our Monte Carlo model, we can calculate the **delta** of each tranche
- Delta is the change in fair premium for a unit change in the premium for the whole portfolio
- To calculate the deltas for each tranche:
  - Run the model at the initial level of default probability
  - Increment the default probability by, say, 0.10%
  - Run the model at the new level of default probability
  - Divide the change in fair premium by the change in the premium for the whole portfolio
- The deltas for the example CDX are shown here:

Tranche	ρ=0	ρ=0.1	ρ=0.2	ρ=0.3	ρ=0.4
0%-3%	3,333%	3,077%	2,550%	2,180%	1,890%
3%-7%	0%	174%	492%	591%	640%
7%-10%	0%	21%	102%	191%	266%
10%-15%	0%	2%	13%	81%	109%
15%-30%	0%	0%	1%	7%	29%

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Trading Correlation – The Long/Short Trade

- It is now possible to create a position immune to changes in default probability
  - Short the 0%-3% tranche for a payment of L+15.95%
  - Long the 3%-7% tranche in a ratio of 3.69 times (2180%/591%) for a return of L+8.38%
- If default probability remains at 1.0%, but correlation falls to 0.2, the results are:

	Net Margin	Net Return	Std Dev	Return / Risk
Long/Short	−7.57%	+4.92%	33.17%	0.15

- Now look what happens if default probability increases to 1.5%:

	Net Margin	Net Return	Std Dev	Return / Risk
Long/Short	−7.57%	+6.89%	44.69%	0.15

- The profits are almost unchanged!
- We have therefore created a short correlation trade immune to changes in default probability

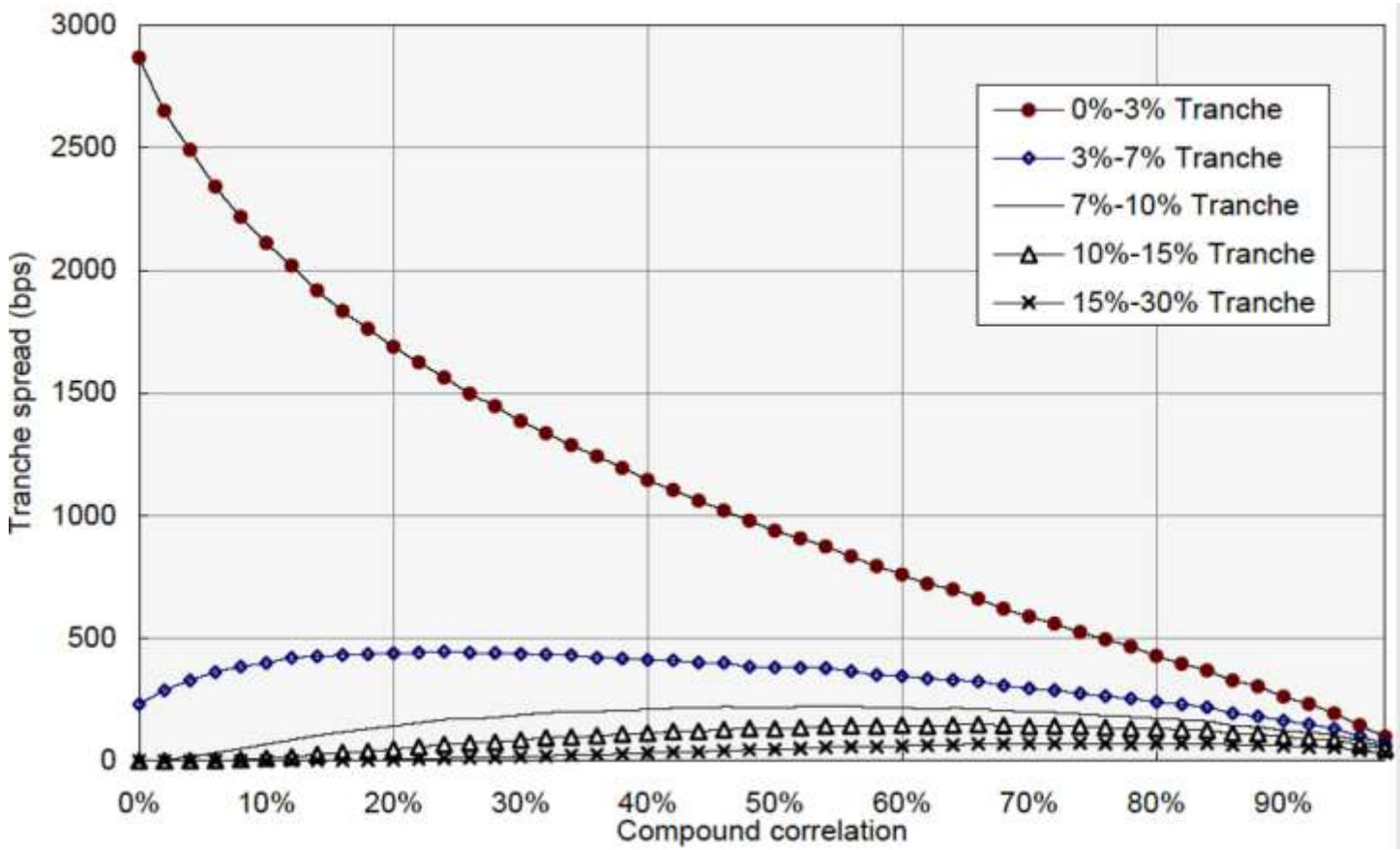
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The Problem with Implied Correlation – 1

Tranche correlation provides sensible results for equity and super senior tranches:

- Equity tranche: higher correlation **lowers** the fair premium
- Super-senior tranche: higher correlation **raises** the fair premium
- For mezzanine tranches the result is ambiguous:
  - Starting with $\rho=0$, increasing correlation **raises** the fair premium…
  - But as $\rho \rightarrow 1$, increasing correlation **reduces** the fair premium
  - There are therefore two values of implied correlation obtainable from one premium!

Compound correlation



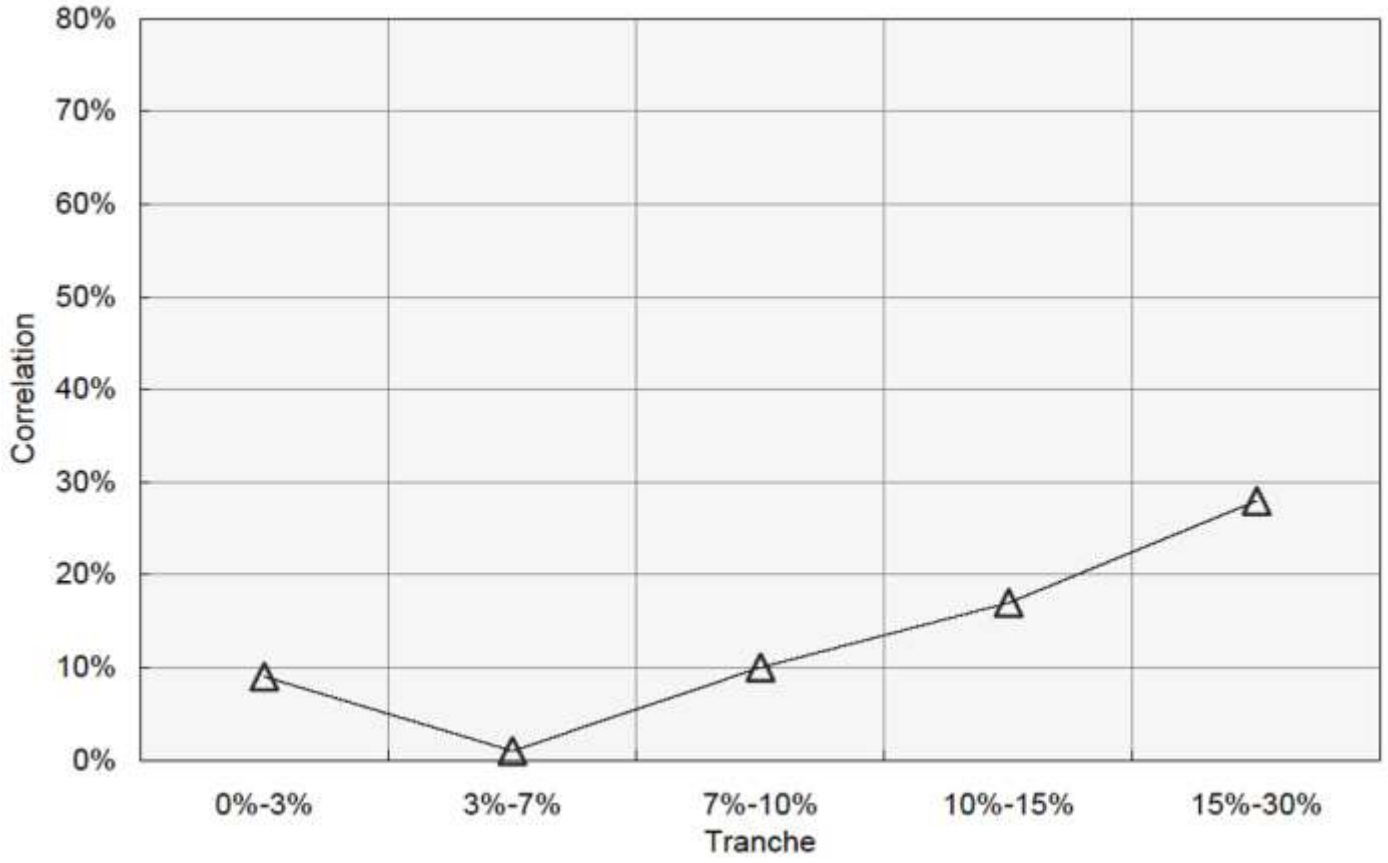
Source: Nomura

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The Problem with Implied Correlation – 2

- Implied correlation also exhibits a ‘smile’ effect
- This ‘smile’ arises because of:
  - Supply and demand
  - Model deficiencies
- The correlation smile makes it difficult to interpolate when pricing a customized tranche with non-standard attachment or detachment points

Compound correlation ‘smile’ of the 5-year CDX.NA.IG Index



Source: Creditflux. September 27 2005

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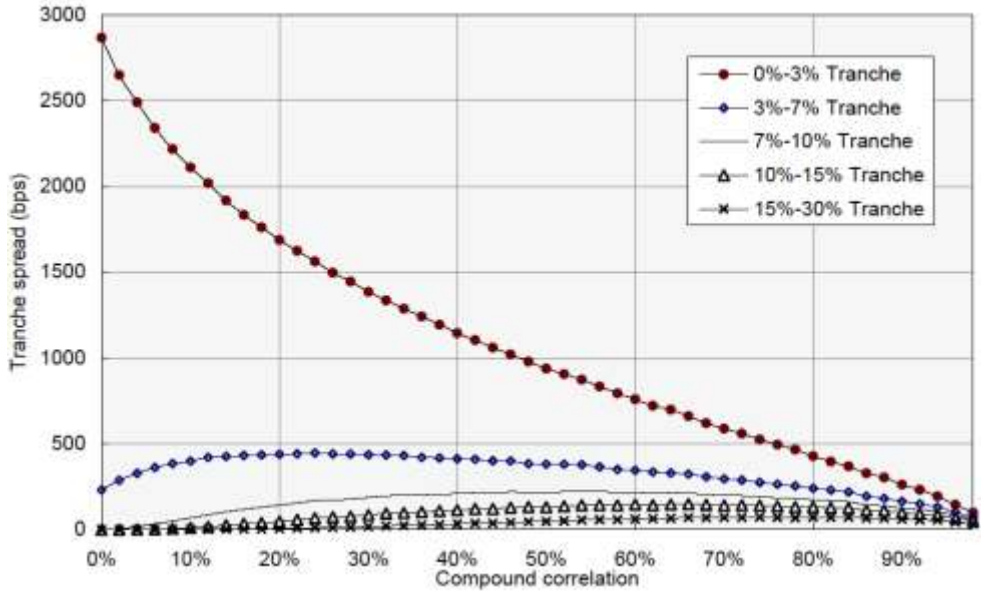
Base Correlation

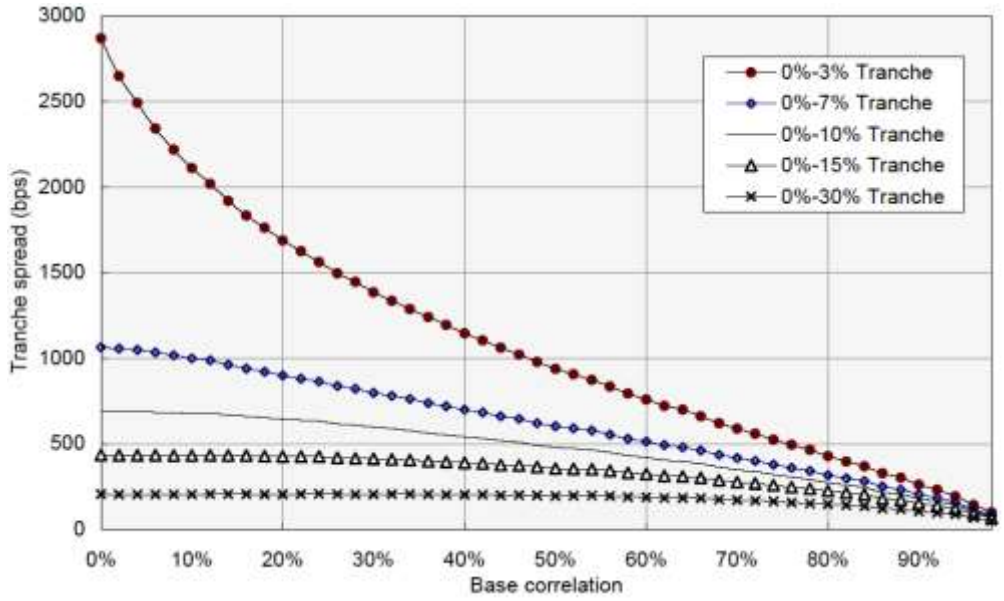
- Base correlation for a tranche is the correlation with an attachment point of 0%

Examples:

- Base correlation for a 0% to 3% tranche is: the implied correlation of the 0%-3% tranche
- Base correlation for a 3% to 7% tranche is: the implied correlation of the 0%-7% tranche
- Base correlation for a 7% to 10% tranche is: the implied correlation of the 0%-10% tranche
- Base correlation for a 10% to 15% tranche is: the implied correlation of the 0%-15% tranche

- This provides a unique value for implied correlation for all tranches…



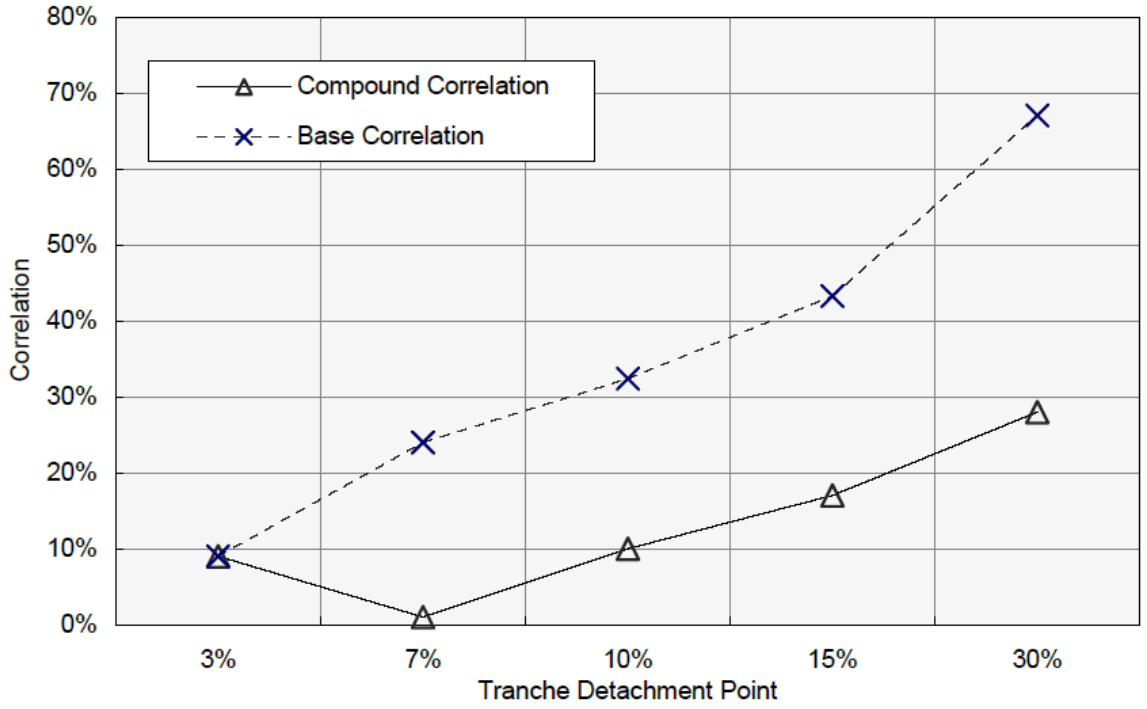


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Advantages of Base Correlation

- Provides unique implied correlation for any market premium
- Interpolation of base correlation easier when pricing non-standard tranches

Compound correlation vs. base correlation (September 27, 2005)



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Disadvantages of Base Correlation

- Base correlation is less intuitive and more difficult to interpret than compound correlation
- You need to know the spread levels of **all** tranches up to a certain detachment point to calculate base correlation
- Base correlation therefore reflects market spreads for multiple tranches, making it difficult to separate market condition for each tranche

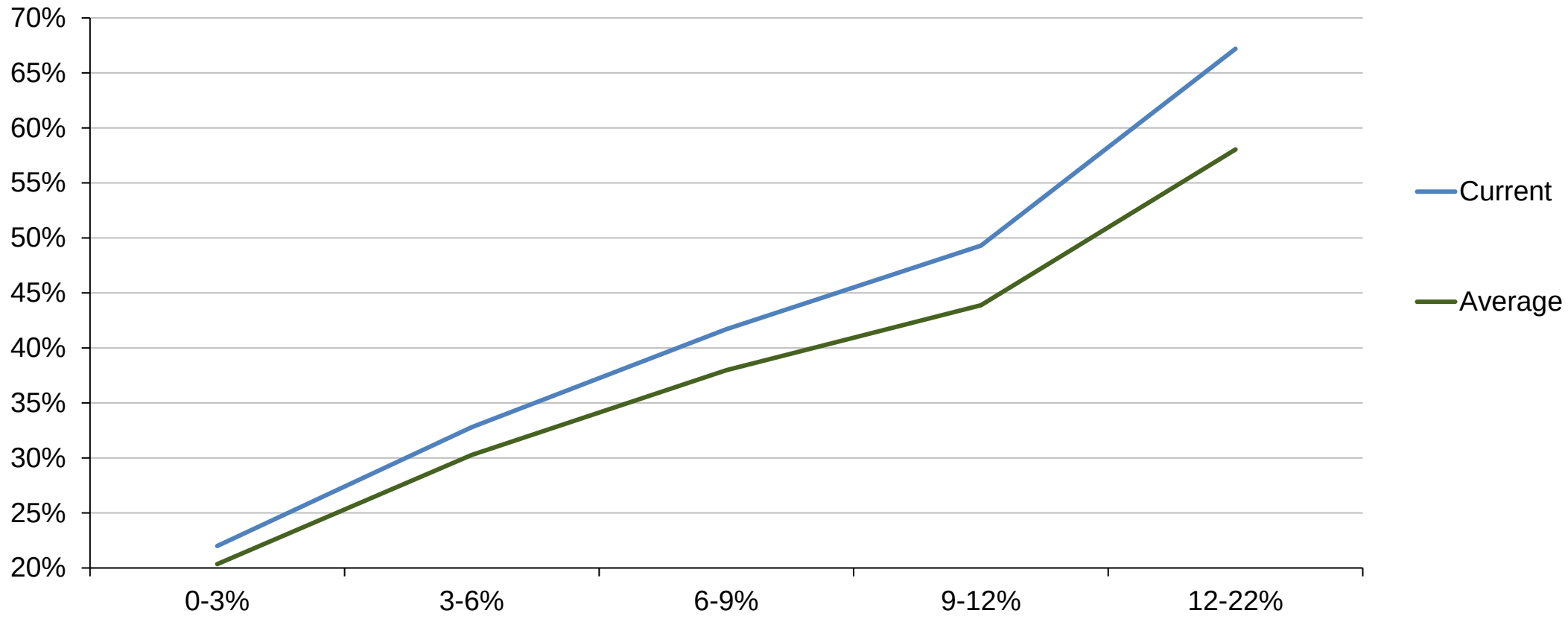
Example

  - If the 0%-3% tranche becomes undervalued ("cheap") while …
  - The 3%-7% tranche becomes overvalued ("rich")…
  - Base correlation for the 0%-7% tranche may reflect the offsetting forces and show little change
- Base correlation does not resolve the correlation smile…
- … the smile is simply transformed into a "skew"

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Example 5-year iTraxx Base Correlation Curve

Base Correlation curves as a function of detachment point



Source: Morgan Stanley as of 27 April 2005

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Risk Management

We saw earlier that the performance of each tranche depends on:

- Default probability – the likelihood of default of each asset in the underlying pool
- Recovery rates – the financial impact of a credit event
- Correlation – the correlation between defaults of each asset in the underlying pool

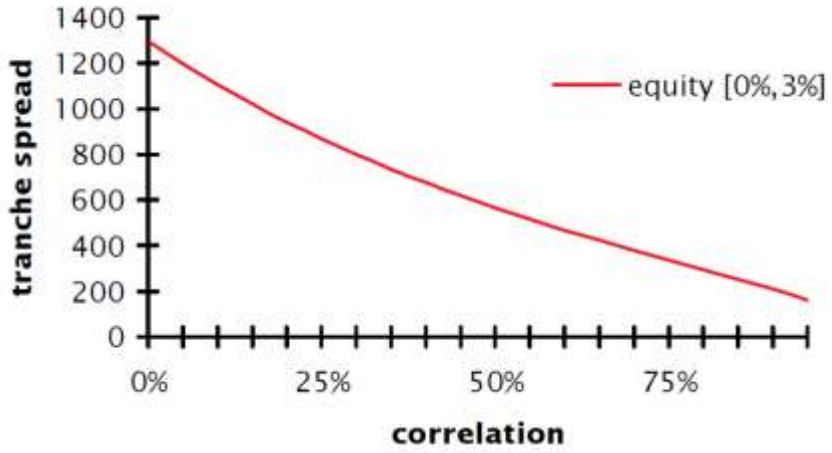
We will now explore these risk factors in greater depth, focussing on:

- Correlation risk
- Spread convexity
- Recovery rate risk
- JTD risk
- Time decay

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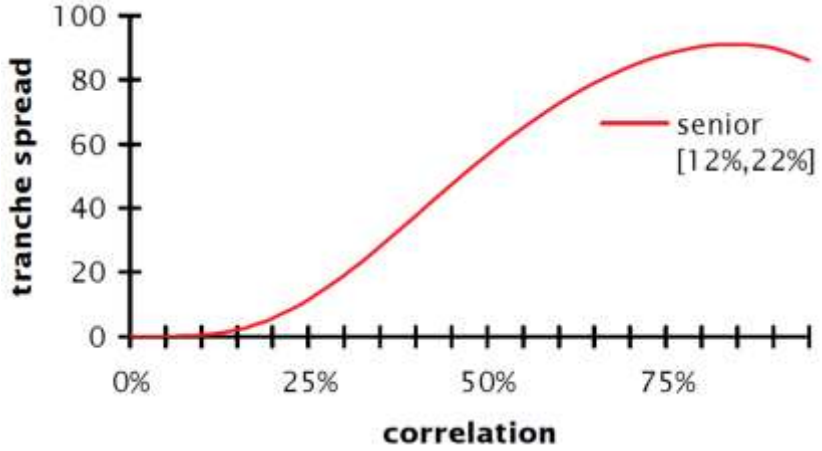
Correlation Risk – Equity and Senior Tranches

Equity spread tightens when correlation increases



Strikes = [0%, 3%]. Average spread = 43bp, maturity = 5y
Source: SG Credit Research

Senior spread widens when correlation increases



Strikes = [12%, 22%]. Average spread = 43bp, maturity = 5y
Source: SG Credit Research

As seen earlier:

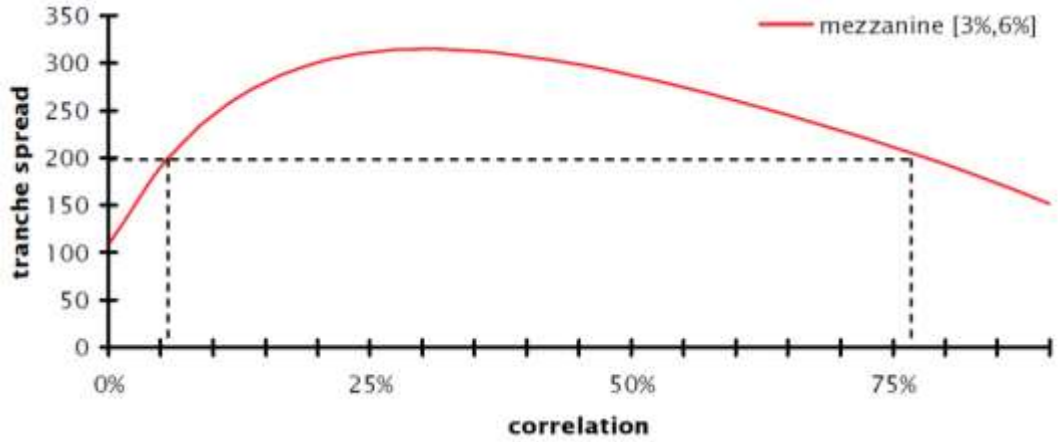
- For the equity tranche, as correlation increases, losses **decrease** (**long** correlation)
  - Multiple credit events are more likely, but equity tranche losses are capped by detachment point
  - Probability of no defaults also increases (significantly); this is what cheapens the equity tranche
- For senior tranches, as correlation increases, losses **increase** (**short** correlation)
  - Multiple credit events are more likely, increasing chance of reaching attachment point

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Correlation Risk – Mezzanine Tranches

- For mezzanine tranches, there is an ambiguous relationship
  - When correlation is low, mezzanine tranches behave more like senior tranches, and losses **increase** as correlation increases
  - When correlation is high, mezzanine tranches behave more like the equity tranche, and losses **decrease** as correlation increases

Mezzanine tranches and correlation



Strikes = [3%, 6%]. Average spread = 43bp, maturity = 5y

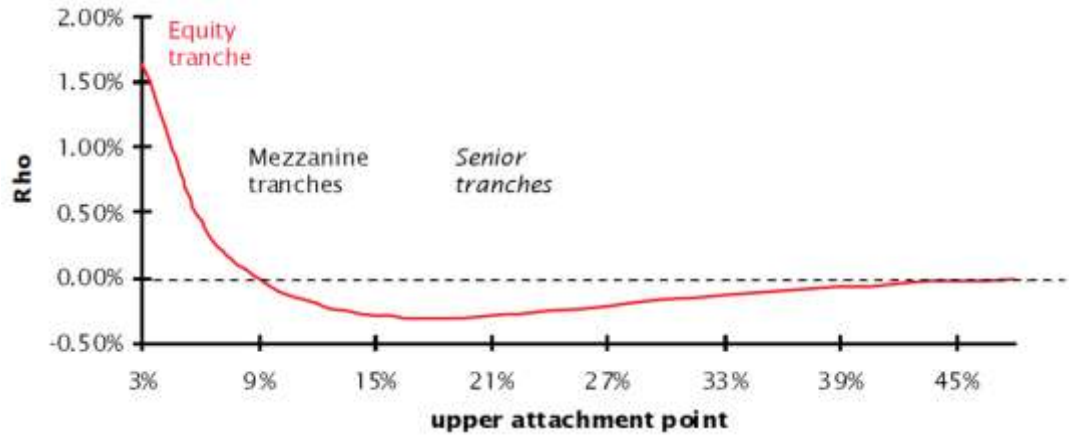
Source: SG Credit Research

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Correlation Risk – Rho

- Correlation sensitivity defined as:
  - Change in mark-to-market value of a tranche, for a [small]
  - Change in the correlation used to price the tranche
- Correlation sensitivity designated as **rho**
- Rho depends on:
  - Subordination of the single-tranche CDX
  - Average level of credit spreads in the underlying portfolio
- The chart opposite shows how rho varies with subordination for a 3% wide tranche

Correlation sensitivity (rho)

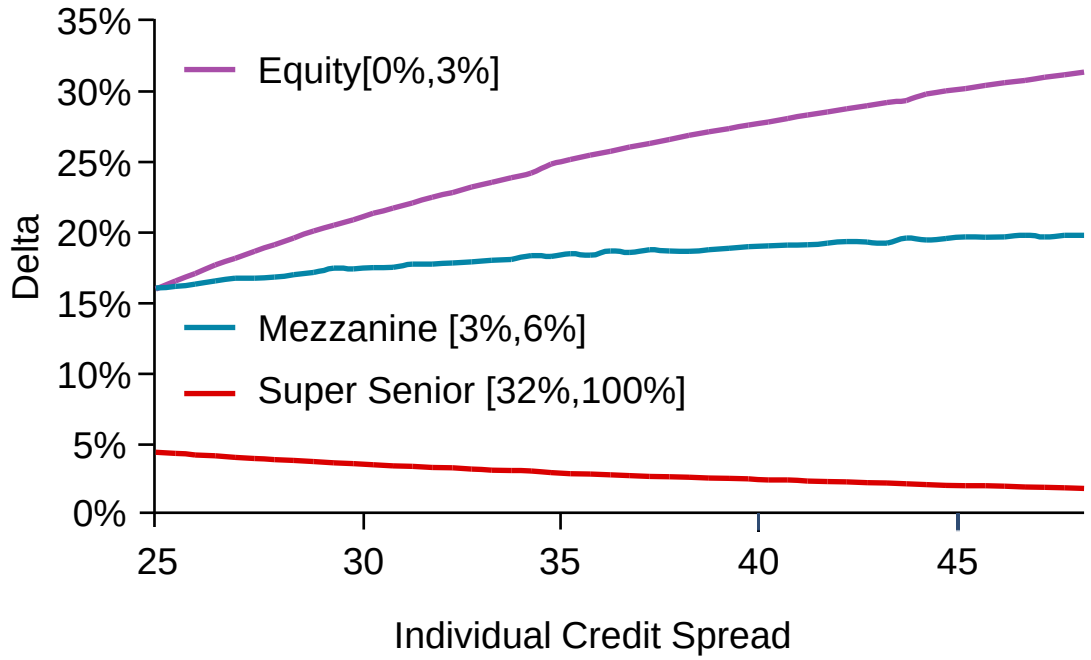


Average credit spread = 50, maturity = 5y. Sensitivity on a long CDO position.
Source: SG Credit Research

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Delta and Individual Credit Spreads

- The chart opposite shows the deltas for a portfolio whose individual credit spreads are uniformly distributed between 25bp and 75bp
- Delta here is defined as:
  - Change in MTM of tranche, for a [small]
  - Change in MTM of an individual name
- As the spread of an **individual name** widens, the implied PD of that name increases, which will increase:
  - The likelihood of the default occurring earlier and hitting the equity tranche
  - The delta for the equity tranche
- … and will decrease the likelihood of the default hitting the senior tranche

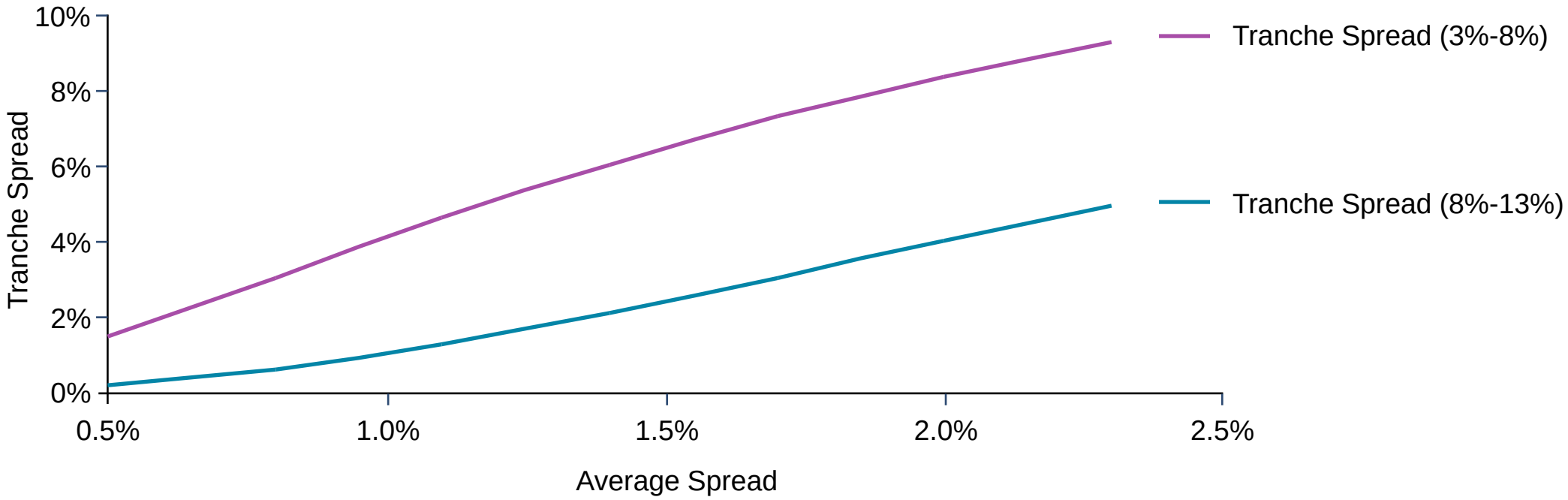


- This leads to an apparent paradox when hedging the senior tranche:
  - A smaller hedge is needed on the riskiest names
  - A larger hedge is needed for the safest names!

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Tranche Spread and Basket Spread

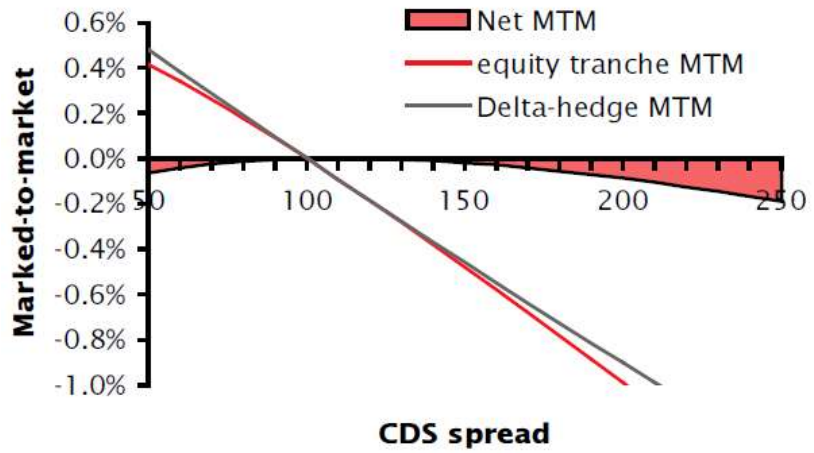
- As may be expected, the wider the average spread on the basket…
- … The wider the spread on the tranches
- Also note that the ratio for the lower tranches is greater than 1:1



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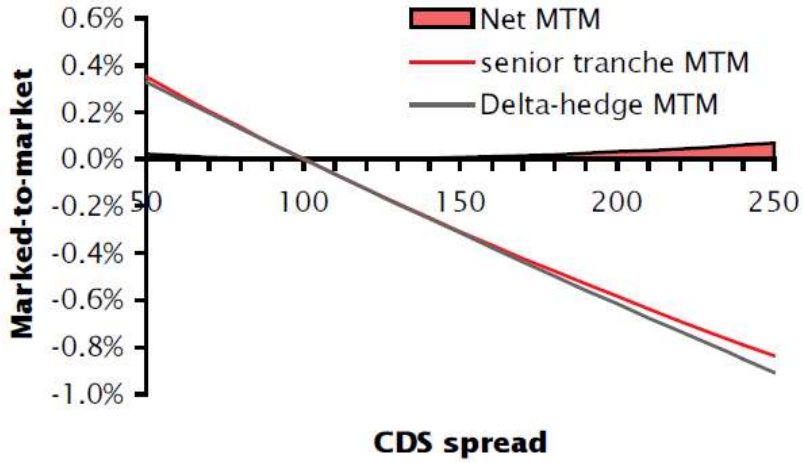
Spread Convexity – Individual Credit Spreads

When one spread widens, the equity tranche exhibits concavity…



Source: SG Credit Research

…whereas the senior tranches benefit from convexity



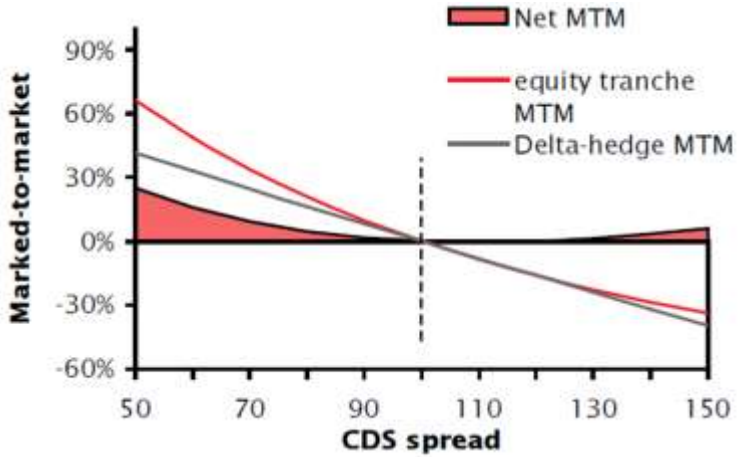
Source: SG Credit Research

- Equity tranche has negative convexity when individual credit spreads widen
  - As seen on the previous pages, wider individual CDS premiums hurt the equity tranche
  - An investor delta-hedging a long position in the equity tranche must:
    - Buy additional protection when CDS spreads are higher
    - Sell protection when CDS spreads are lower
- Senior tranches conversely have positive convexity

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Spread Convexity – Basket Spreads

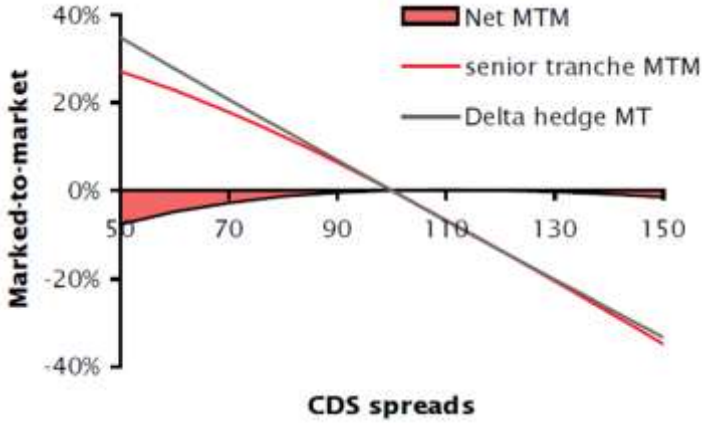
When all the underlying spreads move, the equity tranche benefits from convexity…



All the underlying CDS spreads shift away from their initial spread of 100bp

Source: SG Credit Research

…whereas the senior tranches present a concave profile



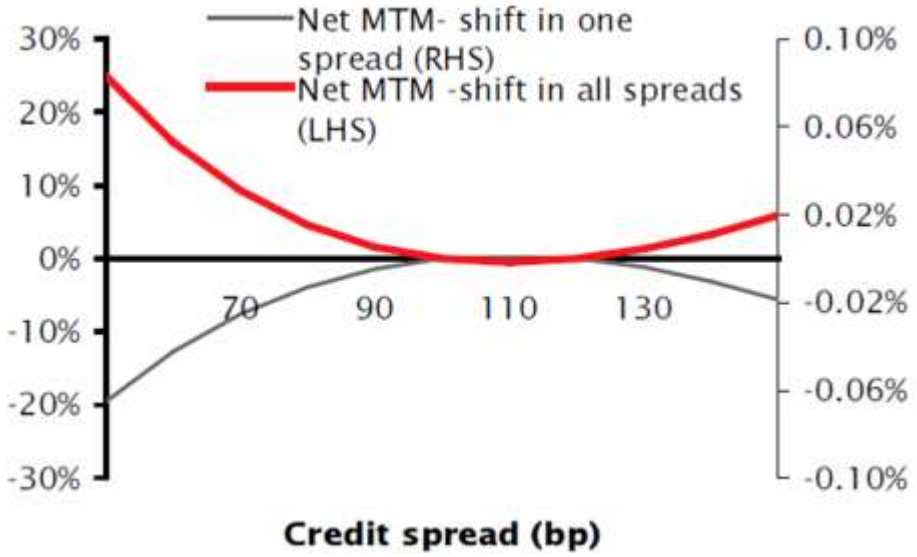
Source: SG Credit Research

- Equity tranche has positive convexity when basket spreads widen
  - As seen on the previous pages, wider overall premiums also hurt the equity tranche
  - But wider overall premiums affect the delta hedge even more, meaning that to maintain delta zero:
    - Equity investor must sell protection when CDS spreads are higher
    - Equity investor must buy more protection when CDS spreads are lower
- Senior tranches conversely have negative convexity

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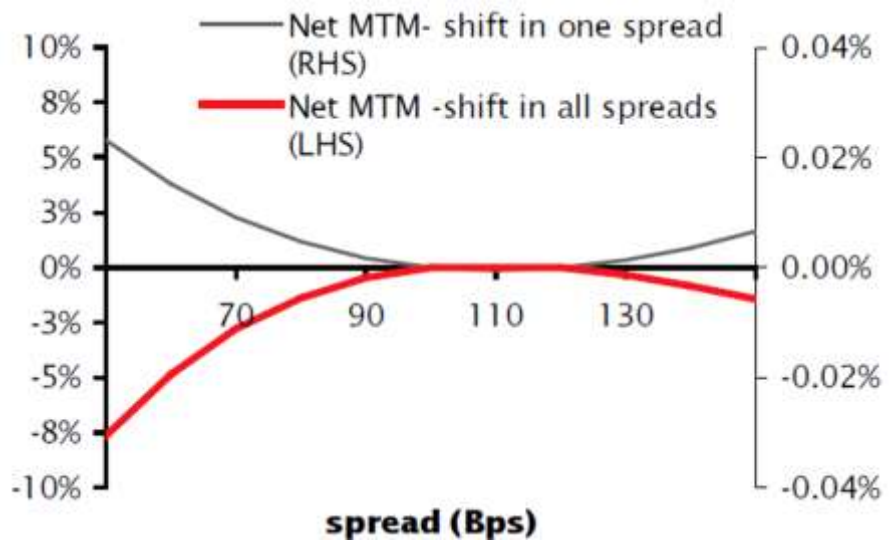
Spread Convexity – P&L Comparisons

Delta hedged long equity tranche P&L



Source: SG Credit Research

Delta hedged long senior tranche P&L



Source: SG Credit Research

- As the charts above illustrate, spread convexity is beneficial for the equity tranche when all credit spreads widen…
- … and detrimental to the senior tranche

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Recovery Rate Risk

- Remember that in the simplest CDS premium model:

$$CDS\ Premium\ =\ PD \times (1\text{-}RR)$$

- Varying the recovery rate while keeping the spread fixed, amounts to changing the default probability of the credit
- The dependence of delta on recovery rates is therefore similar to the dependence of delta on CDS spreads

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JTD Risk

- Jump to Default (JTD) risk arises when:
  - A tranched CDX position is hedged with
  - Holdings of the constituent single-name CDS
- If one name defaults, the notional amount of CDS held will be different from the loss incurred
  - … because deltas are seldom 100%
- This gives rise to a JTD risk which is difficult to avoid
- JTD risk is linked to correlation risk

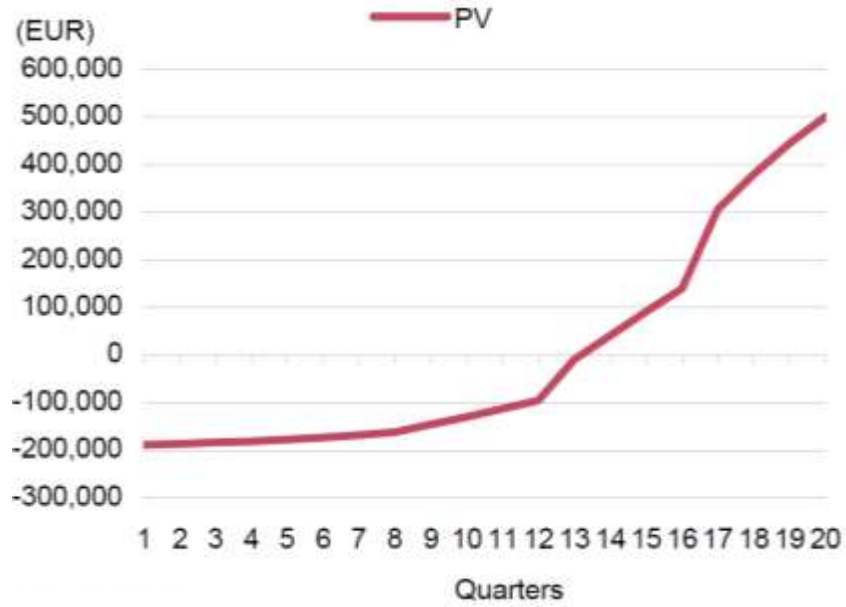
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Time Decay

The profile of expected losses & premium over time



Change in PV per quarter for iTraxx EUR SG 3-6%



- Losses in mezzanine tranches tend to be back-loaded because of subordination
  - Expected losses rise over time
  - Expected premiums will fall over time
- The P&L chart on the right (for the protection buyer) reflects this

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May 2005 Correlation Crisis I

- Real money accounts were heavily selling protection to bank dealers on mezzanine tranches, attracted by:
  - Relative safety from defaults
  - Good returns
- Banks were therefore eager to sell protection on mezzanine tranches to hedge their significant exposures
- Hedge funds and proprietary trading desks found this a good environment to:
  - Sell protection to banks on equity tranches (attracted by their high coupons), and
  - Hedge that by buying protection from banks on mezzanine tranches
- This ‘equity vs. mezz’ trade was extremely popular before May 2005

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May 2005 Correlation Crisis II

- By May 2005, the conditions for GM and Ford were precarious
- On 4 May 2005, billionaire Kirk Kerkorian injected GM shares with their biggest one-day gain in more than 40 years by offering an $870m investment in the company
- On 5 May 2005, however, the rating agency S&P downgraded GM and Ford:
  - GM's $292bn and Ford's $163bn debts became "junk"
- The timing of that decision, coming only one day after Kerkorian's announcement, caught the market off-guard

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May 2005 Correlation Crisis – Index Spreads

- The table below shows the impact of the GM and Ford downgrades on CDX and iTraxx premiums
  - US premiums closed wider on 5 May
  - Europe premiums closed wider on 6 May

    … because the downgrade occurred after European markets had closed
- Credit spreads reached their widest on 17 May 2005
- By 26 May 2005, spreads were back to normal

	4 May	5 May	6 May	17 May	26 May
Eur Main	43.75	41.75	44.25	60.25	42.5
Eur High Vol	84.5	81.5	87	114.5	81
Eur Crossover	320	310	340	447	340
US IG	55.5	62	63.5	78	57
US High Vol	129	143.5	145	175	131

Source: "The May 2005 correlation crisis: Did the models really fail?", Ammar Kherraz, Imperial College London, June 2006

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May 2005 Correlation Crisis – Index Tranches

- The table opposite shows the impact of the downgrade on CDX and iTraxx tranches
- There is evidently an equity-mezzanine dislocation, or simply as a correlation breakdown
- On 26 May 2005:
  - Equity tranches were still wider than the pre-crisis levels
  - Junior mezzanine tranches were significantly tighter
- This rendered the popular “equity vs. mezz” trade a disaster, as both legs of the trade went in the wrong direction

	4 May	5 May	6 May	17 May	26 May
Eur 0–3%	29%	30%	31%	49%	34%
Eur 3–6%	168	158	165	170	122
Eur 6–9%	49	41	41	49	41
Eur 9–12%	25	23	23	34	24.5
Eur 12–22%	16	14	15	24	15
Ref (Eur)	44	43	45	57	44
US 0–3%	44%	46%	47%	62%	51%
US 3–7%	235	229	211	275	185
US 7–10%	55.25	54	47.5	54	51
US 10–15%	28	28	24	30	22
US 15–30%	10.5	11	9	18.5	12.5
Ref (US)	59	60	60	78	60

Source: "The May 2005 correlation crisis: Did the models really fail?", Ammar Kherraz, Imperial College London, June 2006

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